Multimodal VR Interaction Toolkit

Multimodal User Interfaces
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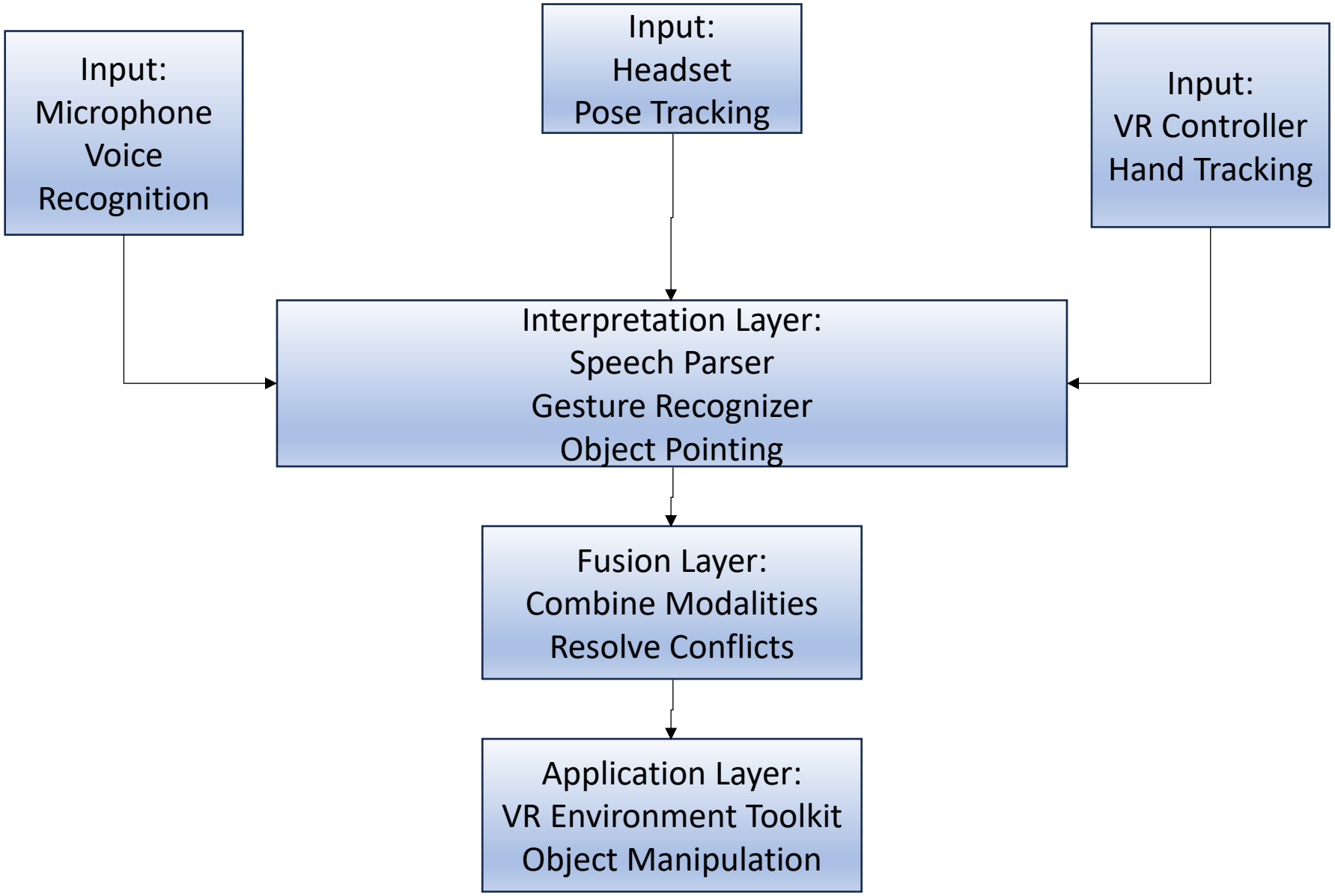
Hands-Free Multimodal VR Interaction

- **Goal**
  Develop a **multimodal VR interaction toolkit** that enables hands-free interaction using **voice commands, gaze direction and motion input(head movement)**.
- **Motivation**
  Traditional VR relies heavily on handheld controllers.
  Our system explores **alternative interaction techniques using voice and eye tracking**.
- **Concept**
  Voice commands trigger actions while **gaze provides spatial context** for selecting and manipulating objects in VR.
- **Example Interactions**
- **Navigation**
  • Look at location + **“Go there”** → teleport
- **Object Selection**
  • Look at object + **“Select this”**
- **Object Manipulation**
  • **“Move this”** → look at location → **“Place here”**

Multimodal VR Interaction Architecture



Thank you for listening!
Any questions?